

Participatory Design of a Serious Game to Improve Wildfire Preparedness with Community Residents and Experts

MJ Johns
Emmanuel Ezenwa Jr.
mljohns@ucsc.edu
eezenwa@ucsc.edu
University of California Santa Cruz
Santa Cruz, California, USA

Seunghyun Lee
seunghyunlee@berkeley.edu
University of California Berkeley
Berkeley, California, USA

Thomas Maiorana
Ciel Wood
tmaiorana@ucdavis.edu
ciewood@ucdavis.edu
University of California Davis
Davis, California, USA

Josh D Levano
Rita Tesfay
Michael Takami
jdlevano@ucsc.edu
rtesfay@ucsc.edu
mtakami@ucsc.edu
University of California Santa Cruz
Santa Cruz, California, USA

Cameron A Dodd
Madison Li
Hanne Manning
cadodd@ucsc.edu
mli254@ucsc.edu
hcmannin@ucsc.edu
University of California Santa Cruz
Santa Cruz, California, USA

Regis Pak
Lily Chen
Ria Saini
rjpak@ucsc.edu
lchen207@ucsc.edu
riksaini@ucsc.edu
University of California Santa Cruz
Santa Cruz, California, USA

Mário Escarce Junior
m.escarce@lancaster.ac.uk
Lancaster University
Lancaster, UK

Mennatullah Hendawy
Magy Seif El-Nasr
mseifeln@ucsc.edu
University of California Santa Cruz
Santa Cruz, California, USA

Edward Melcer
Katherine Isbister
eddie.melcer@ucsc.edu
katherine.isbister@ucsc.edu
University of California Santa Cruz
Santa Cruz, California, USA

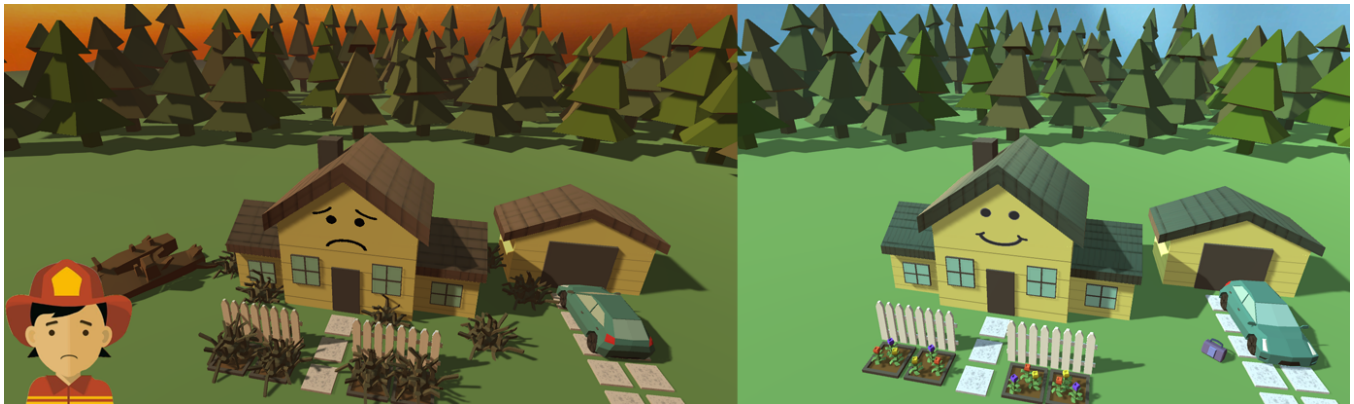


Figure 1: Before and after home-hardening in the Happy House minigame

ABSTRACT

Serious games can be a valuable addition to the resources available for increasing community resilience in the face of disaster. The

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purpose of this project is to work closely with two communities at risk of wildfire to design and develop a serious game that can be used by experts and leaders within those communities as part of their holistic approach to increasing resilience and improving readiness of their community members. We follow a community-centric approach, utilizing Participatory Design and Research through Design to engage directly with community members and local experts. This paper documents our findings at two workshops involving the early game designs and contributes to the broader exploration of the use of Participatory Design with serious games in support of community initiatives around disaster preparedness.

CCS CONCEPTS

• **Human-centered computing;**

KEYWORDS

Serious Game Design, Wildfire Resilience, Climate Change, Community Trust Building

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1 INTRODUCTION

In recent years, climate risks and natural hazards such as wildfires have been on the rise (both in the number of occurrences and the severity of events) and are expected to continue this trajectory [16]. With this trend in mind, we seek community-centric solutions to increase disaster preparedness, improve resilience, and reduce harm (both human and property losses). Our team proposes a community-based serious game to raise awareness, educate, encourage behavior change, and foster communication across impacted communities.

A serious game is a game that exists for purposes beyond entertainment, such as those used for education, training, advocacy, etc [6]. This approach can make complex ideas and instructional material more broadly available and approachable to a wider audience [19]. Serious games have been used for improving resilience in communities, such as the *Kin Dee You Dee* toolkit developed for climate resilience in Thailand [15]. In the case of wildfire preparedness, we hope to use our serious game as a tool to engage with impacted communities to make information about wildfire preparedness more accessible to those who would most benefit from increasing their understanding and awareness, and to motivate them toward specific preparedness actions. Community members themselves are a valuable, yet often underappreciated, resource for preparing for and responding to disasters [21]. Our project process incorporates the community directly into designing the game by inviting community members, leaders, and local experts to participate in design workshops. This Participatory Design approach allows us to generate and curate gameplay that meets the needs and interests of the communities where the finished game will be deployed [24]. Additionally, this allows us to incorporate first-hand accounts of community experiences directly into our design process, and helps us ensure accuracy of the information and procedures presented in the gameplay.

2 MOTIVATION AND CONTRIBUTION

The risk of wildfire and the challenges of prevention and mitigation are complex problems that require community buy-in as well as collaboration across various organizations with nuanced tensions between stakeholders. Our approach incorporates the community members into the design process with a goal of creating a product

that will actively foster conversations within that community as well as building trust at the community level.

While some aspects of our approach to Participatory Design for a serious game addressing wildfire preparedness are new, the general process we are following for developing community-centric gaming experiences has a proven track record. In his book *Locally Played* [2], Benjamin Stokes demonstrates through several case studies the power of games for transformational goals within communities, particularly by leveraging the lived experiences of individuals and their interactions with others in the community.

One specific area of value we hope to bring to the conversation is addressing some challenges of using Participatory Design for games – in [13] the authors note that facilitating player involvement in the design of games can lead to a more challenging development process as players may not have the needed insight and vocabulary to contribute effective game design elements, especially in the case of serious games. We are taking a modular approach – designing a collection of minigames – which enables more flexible community interactions by allowing participants to iterate on our existing designs (via storyboards) or propose an entirely new minigame to add to our collection.

Our contribution with this overall project is to build upon known effective approaches for community-centric game design coupled with the practice of Participatory Design for further community integration, and apply this approach to the significant real-world problem of communities impacted by wildfire risk. In this paper we present results from workshops with stakeholders that have informed our design.

3 RELATED WORK: EXISTING COMMUNITY INTERVENTIONS AND GAMES

Current community preparedness strategies often involve planning forms/instructions (such as the Berkeley Ready Household Fire Weather Planning Tool [4]) and informational websites (such as the one provided by CalFire [23]) as well as various community-specific alert systems like the Alert Marin Emergency Notification System [5] and local events like those hosted by Fire Safe Marin [8]. There have also been some attempts to consider fire in games. One example, *This is Fine* [14], is a game about putting out a fire inside your house, but while it may help to raise awareness it does not offer robust actionable preparation steps.

Beyond wildfire preparation, there are games and interactive narratives addressing climate change and disaster preparedness more broadly, including *Stop Disasters* [18] and the interactive comic *Disaster Master* [20]. In [18] players learn about risks of disasters through realistic simulations, but the harsh penalties and intensity may be discouraging for some players. Conversely, [20] abstracts the concepts to a cartoonish level in the style of a superhero comic book where the lack of realistic scenarios and meaningful feedback may limit the amount of learning taking place.

It is clear from the variety of approaches taken by many different groups and organizations that there is a strong desire at all levels, from grassroots movements to government leadership, to create meaningful content to help communities become more prepared for future disasters. The work done in both categories has helped

to improve community preparedness. However, so long as we continue to experience adverse societal impacts from these disasters we know there is more work to do and room for continued improvement on the current efforts. As a result, we leveraged the insights gained from past work to develop a more targeted and impactful game focused particularly on wildfires and with local specificity to more closely engage with the communities at risk. We believe this approach will make information and instructions more precise and therefore more likely to result in effective strategies being implemented by individuals. As the project progresses, we will expand our reach to the entire region based on what we learn from the initial communities, and from there we may be able to abstract to a point where it can be broadly applicable to many regions while still honing in on individual community needs.

From our examination of existing disaster preparedness games and interactive experiences, we have identified several key components we intend to either embrace or avoid. Games with overly harsh penalties for wrong answers punish players for mistakes and may discourage them from continuing [10]. On the other hand, games that are supportive and encourage trying again can help players feel more empowered and willing to continue and learn from their mistakes [10] [17]. Replayability is also an important aspect for players to achieve proficiency [7]. Another common issue identified in our investigation of prior games was the need for variation in gameplay. I.e. in games aimed at educating or training players, it is important to support replayability in order for the player to have more engagement with the learning content. This is known to be one of the major challenges at the intersection of games and training [7]. Lastly we identified the significance of accessibility features and inclusive design, particularly in consideration of the anticipated audience who may generally have limited prior experience with games [9].

4 PARTICIPATORY DESIGN PROCESS

To gain buy-in at the community level, it is important for researchers to build a level of trust with members of the involved communities. The work discussed in this paper is part of a larger research effort that has involved consistent interaction within the same communities over an extended period of time, which has helped to forge meaningful connections that supported the game design work and allowed this project to take place.

To develop a game that is relevant to these specific communities, we utilized a participatory design process [24] coupled with a research through design (RtD) [11] approach in tandem as we developed the games themselves. Participatory design is a crucial approach for the development of our game since it allows us to generate and curate gameplay that meets the needs and interests of communities where the game will be deployed [24]. Similarly, RtD allows us to iteratively develop our serious game and explore its efficacy in improving the overall preparedness within the communities [1] [25]. This approach is also aligned with most game development processes, which are often iterative [12]. Our participatory design process unfolded as follows:

- (1) Expert Interviews to Identify Needs: two members of the team met with 54 community members and experts and conducted interviews to determine the most pressing needs

for those communities in terms of preparedness for wildfires. The results of the interviews were transcribed and analyzed, and common trends were used to inform the design direction. One particular aspect that emerged was the need to present different personas – people within the community who will experience a wildfire in different ways (such as homeowners, renters, people with livestock, etc.).

- (2) Community Workshops to Solicit Initial Ideas and Feedback: interviewees from the previous section, and other community members and experts, were invited to participate in a series of workshops to discuss initial ideas about the needs that surfaced from the interviews. These workshops were “listening” sessions for the design team to hear directly from individuals in the community about what would help their communities at large.
- (3) Research Team Design Workshop and Prototyping: the research team met for an extended brainstorming session to kick off our design process by generating ideas that combined our collective experience with gamification and our findings from steps one and two about the needs of the community. From this initial brainstorm, we selected nine ideas to write up as design proposals, and four were further developed as prototypes.
- (4) Community Workshops for Feedback and Iteration on Designs and Prototypes: Once we developed the initial design of the overall game concept and several specific minigames, we took our ideas to the community and engaged with community leaders, experts, and residents to iterate and evolve the design. We hosted two workshops, one in each of our chosen communities – which are high risk areas for wildfires. Participants of these workshops were selected through Purposive Sampling (participants actively involved in community wildfire preparation or with expertise or interest in this area). The community stakeholders then participated in a workshop-style brainstorming session where they critiqued and iterated on the ideas as well as offering new ideas for other minigames.

5 GAME DESIGN INSIGHTS FOR COMMUNITY ENGAGEMENT

Using what we have learned from existing interventions, we are developing a serious game intended to be deployed initially in two communities in a west coast state that are known to be at high risk of wildfire impact.

Through the discussions with community leaders and wildfire experts in both locations in steps one and two of our PD process, we have identified two probable scenarios where players are likely to encounter the game we are developing. The first scenario is traditional – players access the game online while at home with family or friends. In this scenario the game must provide the instructional content since it may be the primary source of information for these players.

The second scenario is at a community gathering or formal educational setting, where an expert or facilitator is guiding a larger group through the steps of preparedness and uses the game as a support tool to engage the crowd in active discussion. This is a

novel situation which can benefit from PD strategies to ensure that the hosts of such workshops can use this game effectively as a tool for instruction. In this scenario, the game should leave room for the on-site expert to provide information, which may be very specific to the particular group they are working with (such as information about a city-wide plan presented at a town hall meeting, or evacuation procedures being taught in a school classroom). In these cases, the game should be flexible and fun and offer points of collaboration and discussion that could be broadly applicable. There is less emphasis on presenting specific instruction within the game since this guidance can – and should – be offered by the expert or facilitator.

In both situations, a concern repeatedly brought up by community stakeholders was the need to support a diverse audience with limited gaming experience. The populations impacted may be elderly residents, young families, people experiencing housing insecurity, college students, immigrants, and many others. There are also edge cases to consider like people with large animals such as horses.

6 DESIGN SOLUTION: MINIGAMES

To accomplish the two scenarios mentioned above within the same game, we have taken a minigame approach to the design. The game will exist as a suite of unique minigames that can be accessed individually from a “Hosted Mode” menu to be played in a facilitated session, or encountered in sequence in a “Story Mode” playthrough to be played independently or with friends and family.

In addition to supporting the two specific play scenarios, this approach also supports our need to offer the player a variety of situations to respond to as well as to emphasize specific learning outcomes. Each minigame can be set in a different time and place and with a different character persona at the center of the experience, and with different challenges and objectives for the player to work through.

In order to support a diverse audience with minimal gaming experience, we have chosen mechanics that align with casual mobile and web games (hidden object, point-and-click adventure, interactive narrative). The interface is designed to be user-friendly and accessible with considerations to make it approachable for an older audience (65+) including larger buttons and text and alternate ways to interact using simplified actions.

Past research on instructional games has indicated that game mechanics should be closely tied to the intended learning outcomes. As Camingue et al (2020) describes, introducing challenges in the gameplay can cause extraneous cognitive load, and in the case where it is unrelated to the topics being taught it can lead to reduced learning outcomes [3]. Therefore, we place significant emphasis on all gameplay being tightly related to the real-world scenarios for which we hope to prepare our users.

7 MINIGAME DESIGNS

The eight minigames designed and presented at the first design/iteration workshop included the following (this list was narrowed down to five for the second workshop):

- Info Overload (Fig 2): A first-person view within a car during an evacuation; the players are presented with information

from various sources including family, city alerts, and radio; intended to be played by two players sharing one screen where one player triages incoming information and the other player makes driving decisions like when to turn or brake based on cars in front of them or emergency vehicles coming from behind; emphasises communication and collaboration



Figure 2: Info Overload Gameplay

- Alert the City: A top-down view of a city; the player plays as a decision-maker (such as the mayor) deciding which zones to evacuate or give shelter-in-place orders; meant to convey the challenges of dealing with a city-wide emergency to help build empathy and respect for decision-makers
- Happy House / Happy Home (Fig 3): View of a house and yard with tasks to complete before evacuation (such as clearing debris or placing a ladder near the house, and marking it as “evacuated” before leaving); House reacts to player’s choices via anthropomorphic emotions (faces on the house indicating happy, sad, scared, guilty)



Figure 3: Happy House Gameplay

- Shelter in Place: Quiz-style minigame where the player is presented with a series of locations and needs to select the “safest” spot to shelter as quickly as possible
- Save Your Neighbor: Top-down view of a small neighborhood; player has options to call or text neighbors/relatives to find their location on the map, then tap those locations to go pick them up (optimization strategy is to find out where everyone is then plan an efficient route to pick everyone up, but may get new “surprise” requests during the game)
- Early Warning (Fig 4): A story-based branching narrative where players make decisions based on situations with limited information; presented as conversational dialog between in-game characters



Figure 4: Early Warning Gameplay

- Find Your Things (Fig 5): Hidden-object game with multiple rooms in a house; player taps items within each room to choose to take them in a “go-bag” and evacuates when they believe they have everything they need (or when they run out of time)

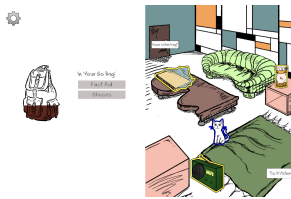


Figure 5: Find Your Things Gameplay

- Smoke Detective (Fig 6): Point-and-click adventure within a house; player taps objects to change their “state” such as closing windows and placing air filters as well as bringing pets inside; goal is to stabilize the on-screen AQI meter in a safe range by sealing their home against smoke



Figure 6: Smoke Detective Gameplay

The minigames were designed with various topics and phases in mind to address the range of issues involved in wildfire preparation. We are working to align the minigames with specific learning objectives following insights from the Transformational Framework [22] and the International Game Developer Association’s Environmental Game Design Playbook [26]. Figure 7 demonstrates how high level thoughts and learning are utilized within specific minigames as we have designed them.

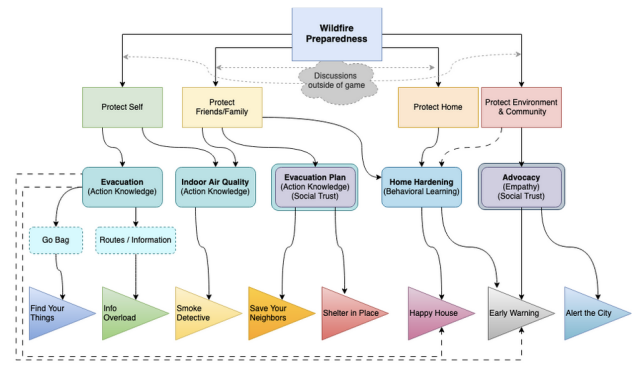


Figure 7: Diagram mapping learning outcomes to minigames

8 PARTICIPATORY WORKSHOP PROCEDURE

We conducted two community workshops (one in each of the communities we are targeting) where participants could engage with our designs contribute to the design process. At each workshop, participants were presented with an overview of the broader design and intent of the project, a playable early prototype of the minigames, video recordings of gameplay, and large printouts of storyboards of each of the minigames. Participants were then encouraged to work in groups and write directly on the storyboards with feedback and ideas (each group was given two minigames to review so they could spend more time focused on specific ideas, rather than trying to provide feedback on all of them). Blank storyboards were also offered to participants who wanted to create their own (at the first workshop we had six participants submit new ideas, and at the second workshop we had three).

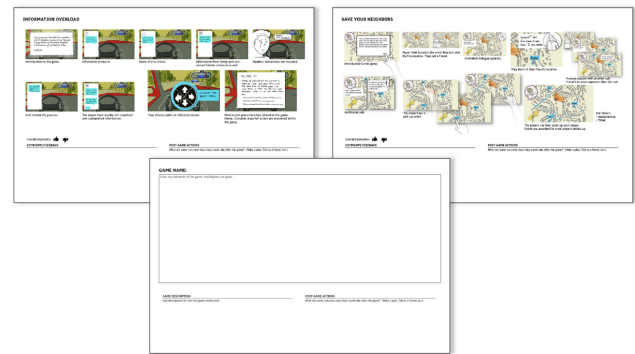


Figure 8: Examples of Storyboard feedback forms used at the workshop

9 PARTICIPATORY WORKSHOP RESULTS

At workshop one, we handed out storyboards for eight minigames. Two of the minigames (Smoke Detective and Shelter in Place) did not receive any responses. Happy House and Early Warning each had four responses, and the remaining minigames received three responses each. At the second workshop we distributed storyboards

for five minigames. Info Overload and Smoke Detective each had one response, Happy House and Early Warning each had three, and Find Your Things had two responses. We collected, typed, and summarized the feedback to share with the design team.

Participants at workshops suggested modifications to our existing designs and narratives, as well as offering additional minigame ideas and character narratives which will be taken into consideration as we develop the games further. Despite some skepticism and apprehension about the use of games for teaching this critical content, they nonetheless offered a great deal of both specific critique of the existing minigame concepts (ie. add flashlight to packing minigame and the mayor is not the right decision-maker for the evacuation minigame) and general concept ideation (ie. show some minigames from a child's or caregiver's perspective and consider a minigame about getting out of a building).

The community feedback and ideation will enable us to iterate on our designs to more closely align with the needs and desires of the people who will be most impacted. The comments and discussion also helped us identify shortcomings in our approach early so that we can adapt and improve moving forward (for example, the need for immediate feedback within the game after each action the player takes, as well as additional perspectives that need to be represented).

Additional details of the results that were gathered during each workshop via written feedback from participants and discussion notes from the investigators can be found in the supplemental materials.

10 DISCUSSION

Based on these findings, we see some key concepts emerging. First is the importance of directing players to reliable data, with a concrete action item to link players to sign up for alerts from reputable sources. Participants of the workshop highlighted this point repeatedly and suggested that community members who do not have access to reliable information or those who receive misinformation will be at high risk during a wildfire and may even cause challenges for others in their vicinity and the emergency responders handling the situation. An additional concern was helping players to decipher whether information is reliable or not when they are making decisions.

The second idea that emerged is the need to consider how these scenarios play out from different perspectives, with a concrete action item of adding a child persona and a caregiver persona to our list of playable characters. It was noted by participants that children can be major drivers of change within communities, and providing them with information relevant to their own perspective could prove extremely useful to the cause.

A third idea highlighted was the importance of giving clear context for each scenario. Participants had some concern about overgeneralizing, when oftentimes a person will need to use situational awareness to make a decision based on the very specific information of the situation they find themselves in. One example provided was deciding where to drive when you see fire ahead – it is not always advisable to drive away from the fire because one may end up driving to an area where they are trapped, so specific situational information must be considered.

Lastly, participants suggested the need to give immediate feedback for in-game actions to help ensure that we are informing and raising awareness effectively and not allowing misconceptions to persist during gameplay.

One observation made by the researchers during and shortly after the workshop was that some of the skepticism about the use of a game for this purpose could be driven by a lack of understanding about the benefits offered by the use of a game. In future presentations, and perhaps within the game itself, we hope to clarify that every individual can benefit from more people in the community having awareness and preparation to respond to the situation, and that games are one tool that can help reach a broader audience and spread critical information to those who might otherwise be unaware.

In terms of broader impact, we hope that this work can serve as a foundation for others who strive to create games and experiences for community impact. The approach of developing minigames can help to encapsulate the critical concepts into distinct pieces of gameplay. By gamifying critical information we can make it more approachable and entertaining, and therefore more likely to spread the ideas within a community.

Engaging early with community members and local leaders and experts can help projects like ours better align with specific needs and interests of the people who will use the resulting product, and potential problems can be caught and addressed to ensure a robust and impactful experience, and Participatory Design is a useful tool in this mission.

Participatory Design has been used effectively in various contexts, but there are challenges specific to integrating it with serious games [13]. Our case study explores its use in the specific communities affected by wildfire, and with particular emphasis on community engagement for improving preparation. One of the unique aspects of this situation is the two distinct play scenarios of individuals/families versus hosted group sessions. Our approach using minigames allows for the flexibility needed to achieve these dual play experiences as well as affording more nuance to the intended learning outcomes desired by the community participants.

11 LIMITATIONS AND FUTURE RESEARCH

The work done up to this point has focused on a limited subset of the community, most with specific skills/interests related to wildfire preparation, and their insights were gathered during just a few hours together in a workshop. As we move into full development of the minigames it will be important to gather broader insights from a larger subset of the community and in settings more aligned with how they would be likely to experience the game.

Additionally, while the approach presented in this paper can help align a project with specific communities, one remaining challenge is how to make this broadly applicable to many communities. It is difficult, perhaps insurmountable, to meet every need of every community that might be impacted. An area for future research might be how to incorporate flexibility into the design and to support emergent gameplay that may differ between communities.

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A WORKSHOP 1 FEEDBACK

During the first workshop, the Serious Games team presented our design concept of a collection of minigames, with several individual minigames demonstrated as prototypes or as storyboards. We gather feedback from the participants during an activity and a discussion, and several highlights are presented below.

Overall Feedback:

- Direct players to reliable information → ie. sign up for alerts
- Consider who is playing and what they need / can do (ex. add child persona)
- Give clear context and background info to setup each minigame (specific and precise)
- Immediate feedback on in-game actions helps to inform and raise awareness

Info Overload Feedback: Incorporate immediate contextual information (ie. stop lights are out, traffic is moving, adapted road behavior, external factors like seeing the fire).

Post-Game Actions: Sign up for alerts; Set communication expectations w/ family

Alert the City Feedback: Logic too simple, consider other decision-makers, not applicable to residents, should replace with a game that builds trust in authorities

Happy Home Feedback: Need reasoning/dialog box with feedback after each item is removed or added; need guidance on how to make the house happier. Post-Game Actions: Have supply setup nearby; Hardening best practices

Early Warning Quiz Game Feedback: Need to add a child persona so they can play too; show who is giving the information / where it is coming from; give the player the option to educate the other person.

Post-Game Actions: Plan for where to go / how long do they stay / where to get more info; Bookmark important websites; Subscribe to emergency text service; Determine contacts

Find Your Things Feedback: Add flashlight; show idea that you might be gone multiple days; give context (ie. red flag day in Marin); make list of last-minute additions; explain when and why you need a go-bag; consider other household members and pets

Save Your Neighbor Feedback: Gives impression people need to play an active role in evacuating others; should be local scale; text works better than phone calls if networks are jammed; consider single family homes vs multi-unit dwellings.

Post-Game Actions: Create a list of contacts who cannot evacuate without help, ie. close vulnerable neighbors; Sign up for alerts / follow instructions from trusted sources

B WORKSHOP 2 FEEDBACK

During the second workshop, the Serious Games team presented our design concept of a collection of minigames, with five individual minigames demonstrated as playable prototypes, videos in the presentation, and as storyboards for participants to write on. We gathered feedback from the participants during the workshop and discussion.

Overall Feedback:

- Direct players to reliable information → ie. sign up for alerts
- Help players identify reliable sources (distinguish unreliable info)

- Consider who is playing and who they help (ex. add caretaker persona)
- Give clear context and instruction to setup each minigame (specific and precise) and more customization (what room do you keep items in)

Find Your Things Feedback: “This concept is great!!”; break into: Go-bag - must be able to carry, life safety for evacuation only → stuff you can prepack | stuff to grab when leaving (phone, wallet, keys); Stuff you want at a shelter *toiletries, medical devices); Sentimental / valuable items (helpful to pre-plan and bring if you have more time); timed game: 1. You have to evacuate RIGHT NOW (evacuation order) grab stuff in 30 seconds; 2. You need to prepare to evacuate (evacuation warning) → grab go bag, shelter stuff, sentimental items; you need to be able to carry your go bag in case you need to get out of your car and run; medical devices; what valuable/sentimental items would they want to bring (could integrate timed test to grab go-bag + last minute items); Does the bag have a limit? is the overall objective of the game for players to ID what they need or to prioritize what they take? Limit size could help prioritize; Reword to less clunky - more direct; conditions to get people thinking

Post-game Actions: Collect critical documents - scan / fire safe; Post game generate personal check list based on questions that people can print/edit; Link to Ready.gov and Ready Set Go; Consequences when you run out of time based on conditions

Happy House Feedback: Offer many choices for setup, e.g. rural home with fence and gate; get residents a little worried about their preparations or lack of preparation; consider neighbors in preparing your house and yard as well as communicating to them; clothing to wear, how to prepare - does that include communication; defensible space (things to do a season ahead) and home hardening 100 ft clearance from home; these actions should be done months ahead

Post-game Actions: outreach to most vulnerable population who may not have access or want to play the game; involve neighbors; join fire safe council for your community

Early Warning Feedback: Create a countdown or multiple countdowns + other decisions; realistically social media is where people would get the instructions (rather than TV); how to assess the credibility / what if information is contradictory; what if you see fire evidence + TV says to stay home → Judgement; user should be guided about when to evacuate vs prepare for evacuation; use steps to be taken before moving on to the next stage rather than options;

Post-game Actions: sign-up to evacuation / safety local notifications; connect to Genasys evacuation notification system

Info Overload Feedback: 2-player element is good - practice working together → but what about for someone alone? Is it 2 players on 2 devices? Depending on skill level in communication of navigator, could be frustrating experience; more likely to stress people out and make them shut down; might not be able to get good directions in an evacuation; a lot of this scenario will be out of the players' control in real life; our org is probably not going to use radio much; nice realistic visual; players will have an unrealistic expectation of being able to get out or be really frustrated and shut down

Smoke Detective Feedback: Maybe add to the game - what air circulation devices are okay to turn on; Start w/ story to set the stage for why they are looking at air quality; ask about pre-existing conditions that could be impacted by air; add info about AQI - what it is and health impacts; ensure this is spelled out; “vent” - specify vents that remove bad air from home

Post-game Actions: Link purple air website; Consider add resources for air filtration or if at high risk - know what conditional resources you may need ie. extra meds; add a warning about seeking medical attention if needed